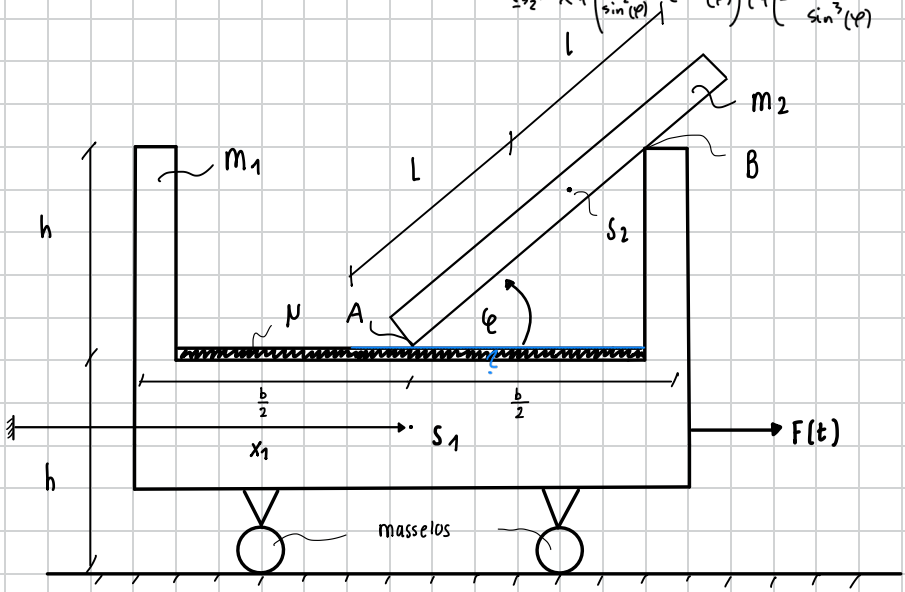
# Aufg. 1 Schwerpunktsatz - Drallsatz

$$\underline{r}_{S_2} = x_1 + \frac{b}{2} - \frac{h}{\tan(\varphi)} + l\cos(\varphi)$$

$$\ddot{\underline{r}}_{S_2} = \ddot{x} + \left(\frac{h}{\sin^2(\varphi)} - l\sin(\varphi)\right)\ddot{\varphi} + \left(-\frac{2h\cos(\varphi)}{\sin^3(\varphi)} - l\cos(\varphi)\right)$$



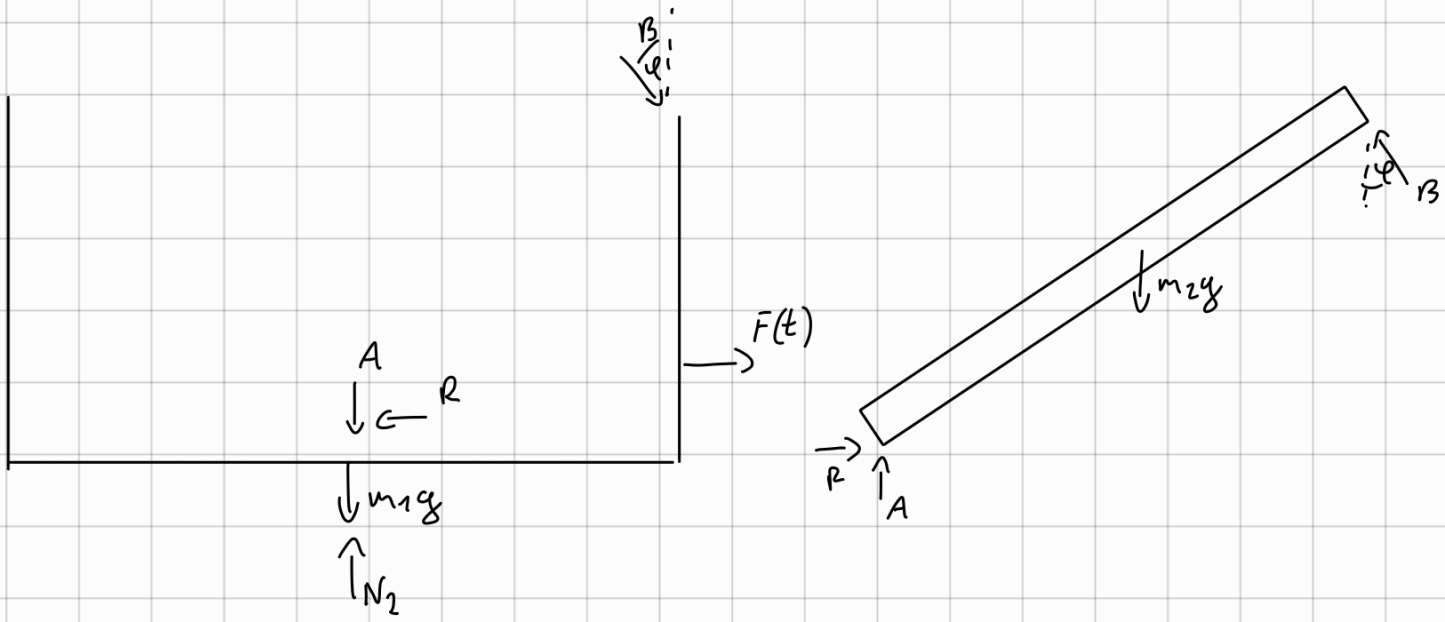

a) ges. $\underline{r}_{S2}$, $\dot{\underline{r}}_{S2}$, $\ddot{\underline{r}}_{S2}$ in Abhängigkeit von $x_1, \varphi$.

b) Freischnitt Stab und Wagen einmal mit Reibung, einmal mit Haftung (mit $\varphi = \pi/4$ gegeben).

c) Wie groß $F_{max}$, damit Haftung vorliegt?

**Lösung a)**

$$\underline{r}_{S2} = \left(x_1 + \frac{b}{2} - \frac{h}{\tan\varphi} + l\cos\varphi\right)\underline{e}_x + \left(h + l\sin\varphi\right)\underline{e}_y$$



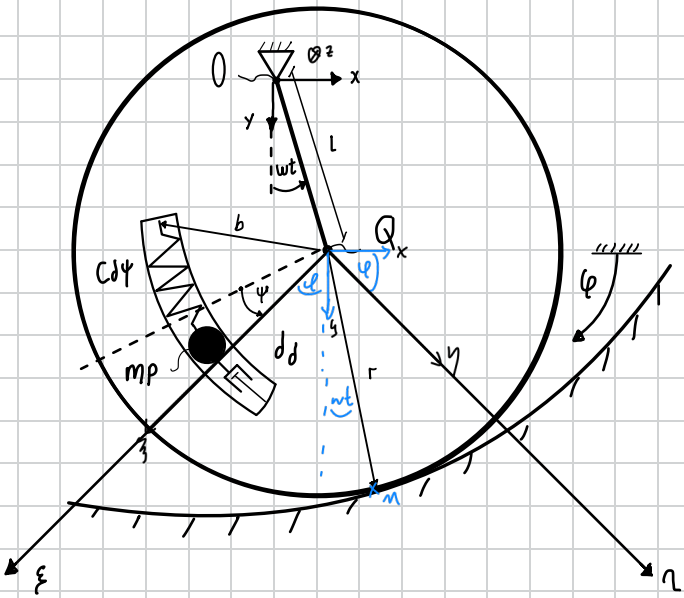

$m_1 \ddot{x}_1 = -R + B \cdot \sin(\varphi) + F(t)$

$m_2 \ddot{y}_2 = N_2 - m_1 g - A - B \cdot \cos(\varphi)$

$m_2 \ddot{x}_2 = R - \sin(\varphi) B$

$m_2 \ddot{y}_2 = A - m_2 g + \cos(\varphi) \cdot B$

# Aufg. 2 Relativmechanik




a) bestimme $\dot{\varphi}(\omega)$, $\varphi(\omega,t)$ in Abhängigkeit von $\omega t$.

Lösung: $\dot{\varphi} = \omega \frac{L}{r}$

b) bestimme $\underline{v}_Q$, $\underline{a}_Q$ und $\underline{r}_{QP}$ im $(\xi,\eta,\zeta)$-System.

c) bestimme $\underline{v}_{führ}$, $\underline{v}_{rel}$ sowie $\underline{v}$.

d) bestimme $\underline{a}_{führ}$, $\underline{a}_{rel}$, $\underline{a}_{cor}$ sowie $\underline{a}$.

e) Schneide die Punktmasse $m_P$ frei und stelle den Schwerpunktsatz in $\xi$ und $\eta$ Richtung auf.

a)

$$\underline{v}_Q = \underline{v}_M + \begin{pmatrix} 0 \\ 0 \\ \dot{\varphi} \end{pmatrix} \times \begin{pmatrix} -r\sin(\omega t) \\ -r\cos(\omega t) \\ 0 \end{pmatrix} = \begin{pmatrix} r\dot{\varphi}\cos(\omega t) \\ -r\dot{\varphi}\sin(\omega t) \\ 0 \end{pmatrix}$$

$$\underline{v}_Q = \underline{v}_O + \begin{pmatrix} 0 \\ 0 \\ -\omega \end{pmatrix} \times \begin{pmatrix} \ell\sin(\omega t) \\ \ell\cos(\omega t) \\ 0 \end{pmatrix} = \begin{pmatrix} \ell\omega\cos(\omega t) \\ -\ell\omega\sin(\omega t) \\ 0 \end{pmatrix}$$

$$\underline{v}_Q \overset{!}{=} \underline{v}_Q \Rightarrow \quad r\dot{\varphi}\cos(\omega t) = \ell\omega\cos(\omega t)$$

$$\dot{\varphi} = \frac{\ell\omega}{r}$$

$$\varphi = \frac{\ell\omega t}{r}$$

b)

trans:

$$\underline{e}_\xi = -\sin(\varphi)\,\underline{e}_x + \cos(\varphi)\,\underline{e}_y$$
$$\underline{e}_\eta = \cos(\varphi)\,\underline{e}_x + \sin(\varphi)\,\underline{e}_y$$
$$\underline{e}_\zeta = -\underline{e}_z$$

✓

$$\underline{\xi}_{QP} = \begin{pmatrix} b\cos(\psi) \\ b\sin(\psi) \\ 0 \end{pmatrix}$$

$$\underline{r}_Q^{x,y,z} = \begin{pmatrix} \ell\sin(\omega t) \\ \ell\cos(\omega t) \\ 0 \end{pmatrix} \qquad \underline{v}_Q = \begin{pmatrix} \ell\omega\cos(\omega t) \\ -\ell\omega\sin(\omega t) \\ 0 \end{pmatrix} \qquad \underline{a}_Q = \begin{pmatrix} -\ell\omega^2\sin(\omega t) + \ell\dot{\omega}\cos(\omega t) \\ -\ell\omega^2\cos(\omega t) - \ell\dot{\omega}\sin(\omega t) \\ 0 \end{pmatrix}$$

in $\xi, \eta, \zeta$

$$\underline{v}_Q = \begin{pmatrix} -\sin(\varphi)\,\ell\omega\cos(\omega t) - \cos(\varphi)\,\ell\omega\sin(\omega t) \\ \cos(\varphi)\,\ell\omega\cos(\omega t) - \sin(\varphi)\,\ell\omega\sin(\omega t) \\ 0 \end{pmatrix}$$

$$\underline{a}_Q = \begin{pmatrix} \left[-\ell\omega^2\sin(\omega t) + \ell\dot{\omega}\cos(\omega t)\right]\sin(\varphi) + \left[-\ell\omega^2\cos(\omega t) - \ell\dot{\omega}\sin(\omega t)\right]\cos(\varphi) \\ \left[-\ell\omega^2\sin(\omega t) + \ell\dot{\omega}\cos(\omega t)\right]\cos(\varphi) + \left[-\ell\omega^2\cos(\omega t) - \ell\dot{\omega}\sin(\omega t)\right]\sin(\varphi) \\ 0 \end{pmatrix}$$

✓

c)

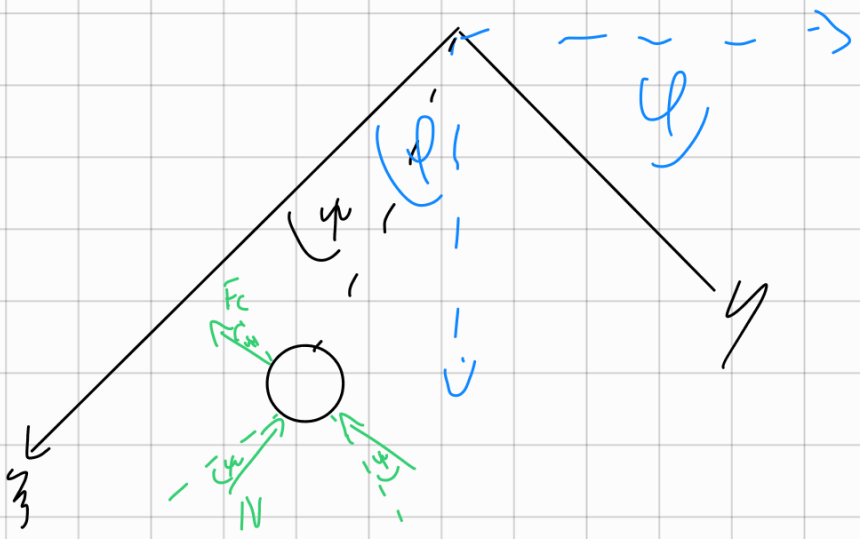
$$\underline{v}_{rel} = \begin{pmatrix} -b\dot{\psi}\sin(\psi) \\ b\dot{\psi}\cos(\psi) \\ 0 \end{pmatrix} \qquad \underline{\omega} \times \underline{\xi}_{QP} = \begin{pmatrix} 0 \\ 0 \\ -\dot{\varphi} \end{pmatrix} \times \begin{pmatrix} b\cos(\psi) \\ b\sin(\psi) \\ 0 \end{pmatrix} = \begin{pmatrix} +b\dot{\varphi}\sin(\psi) \\ -b\dot{\varphi}\cos(\psi) \\ 0 \end{pmatrix}$$

$$\underline{v}_{führ} = \underline{v}_Q + \underline{\omega} \times \underline{\xi}_{QP} = \begin{pmatrix} -\sin(\varphi)\,\ell\omega\cos(\omega t) - \cos(\varphi)\,\ell\omega\sin(\omega t) + b\frac{\ell\omega}{r}\sin(\psi) \\ \cos(\varphi)\,\ell\omega\cos(\omega t) - \sin(\varphi)\,\ell\omega\sin(\omega t) - b\frac{\ell\omega}{r}\cos\psi \\ 0 \end{pmatrix}$$

d)

$$\underline{a}_{rel} = \begin{pmatrix} -b\dot{\varphi}^2\cos(\varphi) - b\ddot{\varphi}\sin(\varphi) \\ -b\dot{\varphi}^2\sin(\varphi) + b\ddot{\varphi}\cos(\varphi) \\ 0 \end{pmatrix}$$

$$\underline{a}_{cor} = 2\underline{\omega} \times \underline{v}_{rel} = \begin{pmatrix} 0 \\ 0 \\ -2\dot{\varphi} \end{pmatrix} \times \begin{pmatrix} -b\dot{\varphi}\sin(\varphi) \\ b\dot{\varphi}\cos(\varphi) \\ 0 \end{pmatrix} = \begin{pmatrix} 2\frac{w_P}{2}b\dot{\varphi}\cos(\varphi) \\ 2\frac{w_P}{2}b\dot{\varphi}\sin(\varphi) \\ 0 \end{pmatrix}$$



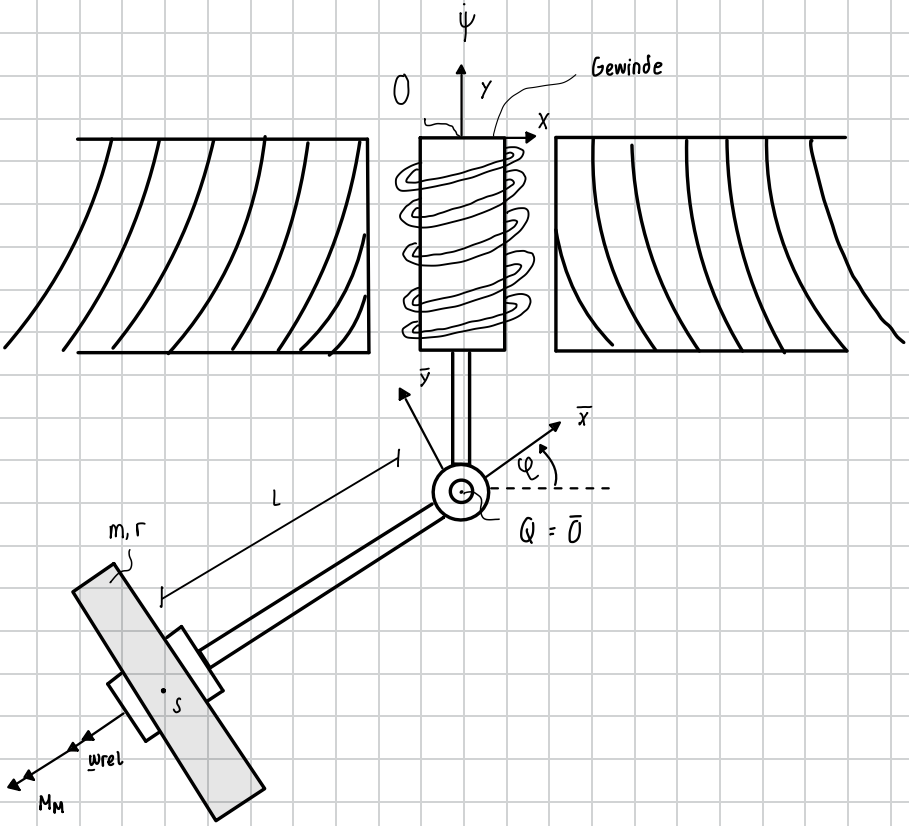

$$F_{cd} = c_{d\varphi} \cdot b \cdot \varphi$$

$$F_{dd} = d_d \cdot b\dot{\varphi}$$

$$m_p \ddot{z} = -N \cdot \cos(\varphi) + c_{d\varphi} b\varphi \cdot \sin(\varphi) + d_d \cdot b \cdot \dot{\varphi} \cdot \sin(\varphi)$$

$$m_p \ddot{y} = -N \cdot \sin(\varphi) - c_{d\varphi} b\varphi \cos(\varphi) - d_d \cdot b\dot{\varphi}\cos(\varphi)$$

# Aufg. 3 Kreisel




Geg: $\dot{y} = h\dot{\psi}$

a) bestimme $\underline{\Omega}$, $\underline{\omega}$ ($\varphi$ und $\Psi$ ins $\bar{x}, \bar{y}, \bar{z}$ System transformieren)

b) bestimme $\underline{a}_Q$ im $(\bar{x}, \bar{y}, \bar{z})$ System.

c) bestimme $\underline{a}_S$ im $(\bar{x}, \bar{y}, \bar{z})$ System.

$\underline{r}_Q = h(\psi - \psi_0)\,\underline{e}_Y$
$\underline{v}_Q = h\dot{\psi}\,\underline{e}_Y$ $\underline{e}_Y$ ins
$\underline{a}_Q = h\ddot{\psi}\,\underline{e}_Y$ $\bar{x}, \bar{y}, \bar{z}$ System

d) Schneide Scheibe frei. Schwerpunktsatz bzgl. S und Drallsatz bzgl. Q

e) Schneide Rahmen frei. Momentensatz bzgl. Q.

trans:

a) $\underline{e}_{\bar{x}} = \cos(\varphi)\,\underline{e}_x + \sin(\varphi)\,\underline{e}_y$

$\underline{e}_{\bar{y}} = -\sin(\varphi)\,\underline{e}_x + \cos(\varphi)\,\underline{e}_y$

$$\underline{\Omega} = \begin{pmatrix} \sin(\varphi)\cdot\dot{\psi} \\ \cos(\varphi)\cdot\dot{\psi} \\ \dot{\varphi} \end{pmatrix} \qquad \dot{\underline{\Omega}} = \begin{pmatrix} \sin(\varphi)\cdot\ddot{\psi} \\ \cos(\varphi)\cdot\ddot{\psi} \\ \ddot{\varphi} \end{pmatrix}$$

$$\underline{\omega} = \begin{pmatrix} \sin(\varphi)\cdot\dot{\psi} - \omega_c\, e \\ \cos(\varphi)\,\dot{\psi} \\ \dot{\varphi} \end{pmatrix}$$

b)

$$\underline{a}_Q = \begin{pmatrix} \sin(\varphi)\cdot h\ddot{\psi} \\ \cos(\varphi)\cdot h\ddot{\psi} \\ 0 \end{pmatrix}$$

c)

$$\underline{a}_S = \underline{a}_Q + \dot{\underline{\Omega}} \times \underline{s}_{QS} + \underline{\Omega} \times \left(\underline{\Omega} \times \underline{s}_{QS}\right)$$

$$\underline{s}_{QS} = \begin{pmatrix} -e \\ 0 \\ 0 \end{pmatrix}$$

$$\underline{\Omega} \times \underline{s}_{QS} = \begin{pmatrix} \sin(\varphi)\cdot\dot{\psi} \\ \cos(\varphi)\cdot\dot{\psi} \\ \dot{\varphi} \end{pmatrix} \times \begin{matrix} -e \\ 0 \\ 0 \\ -e \\ 0 \\ 0 \end{matrix} \quad \begin{matrix} s \\ c \\ \dot{\varphi} \end{matrix} = \begin{pmatrix} 0 \\ -e\dot{\varphi} \\ e\cos(\varphi)\cdot\dot{\psi} \end{pmatrix}$$

$$\dot{\underline{\Omega}} \times \underline{s}_{QS} = \begin{pmatrix} 0 \\ -e\ddot{\varphi} \\ e\cos(\varphi)\cdot\ddot{\psi} \end{pmatrix}$$

$$\underline{\Omega} \times \left(\underline{\Omega} \times \underline{s}_{QS}\right) = \begin{pmatrix} \sin(\varphi)\cdot\dot{\psi} \\ \cos(\varphi)\cdot\dot{\psi} \\ \dot{\varphi} \end{pmatrix} \times \begin{pmatrix} 0 \\ -e\dot{\varphi} \\ e\cos(\varphi)\cdot\dot{\psi} \end{pmatrix} = \begin{pmatrix} e\cos^2(\varphi)\,\dot{\psi}^2 + e\dot{\varphi}^2 \\ -\sin(\varphi)\cdot\dot{\psi}^2 e\cos(\varphi) \\ -\sin(\varphi)\cdot\dot{\psi}\, e\dot{\varphi} \end{pmatrix} \quad \begin{matrix} s \\ c \\ \dot{\varphi} \\ 0 \\ -e\dot{\varphi} \\ e \end{matrix}$$

$$\underline{a}_S = \begin{pmatrix} \sin(\varphi)\cdot h\ddot{\psi} \\ \cos(\varphi)\cdot h\dot{\psi} \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -e\ddot{\varphi} \\ e\cos(\varphi)\cdot\ddot{\psi} \end{pmatrix} + \begin{pmatrix} e\cos^2(\varphi)\,\dot{\psi}^2 + e\dot{\varphi}^2 \\ -\sin(\varphi)\cdot\dot{\psi}^2 e\cos(\varphi) \\ -\sin(\varphi)\cdot\dot{\psi}\, e\dot{\varphi} \end{pmatrix}$$

d)




$$\begin{pmatrix} -l \\ 0 \\ 0 \end{pmatrix} \times m \begin{pmatrix} \sin(\varphi)\cdot h\ddot{\varphi} \\ \cos(\varphi)\cdot h\ddot{\varphi} \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -l\cos(\varphi)h\ddot{\varphi} \end{pmatrix}$$

$$\begin{matrix} -l \\ 0 \\ 0 \end{matrix} \quad \begin{matrix} s \\ c \\ 0 \end{matrix}$$

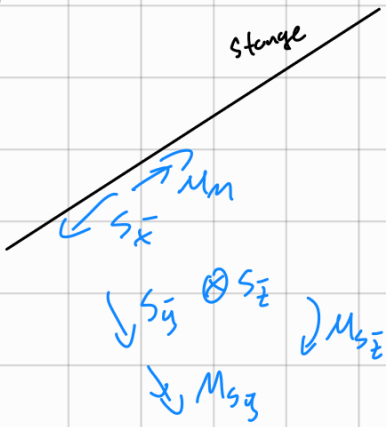
$$m a_{S\bar{x}} = S_{\bar{x}} - mg\sin(\varphi)$$
$$m a_{S\bar{y}} = S_{\bar{y}} - mg\cos(\varphi)$$
$$m a_{S\bar{z}} = S_{\bar{z}}$$

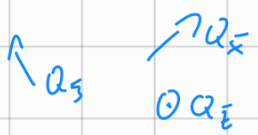
$$\Theta a_{x\bar{x}}\,\dot{\omega}_{\bar{x}} + \begin{pmatrix} \Omega_{\bar{x}} \\ \Omega_{\bar{y}} \\ \Omega_{\bar{z}} \end{pmatrix} \times \begin{pmatrix} \Theta a_{\bar{x}\bar{x}}\omega_{\bar{x}} \\ \Theta a_{\bar{y}\bar{y}}\omega_{\bar{y}} \\ \Theta a_{\bar{z}\bar{z}}\omega_{\bar{z}} \end{pmatrix} \qquad (\underline{S}_S - \underline{S}_a) = \begin{pmatrix} -l \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned}
&\Theta a_{x\bar{x}}\,\dot{\omega}_{\bar{x}} + \left(\Theta a_{\bar{z}\bar{z}}\,\omega_{\bar{z}}\,\Omega_{\bar{y}} - \Omega_{\bar{z}}\,\Theta a_{\bar{y}\bar{y}}\,\omega_{\bar{y}}\right) && = M_{a\bar{x}} \\
&\Theta a_{\bar{y}\bar{y}}\,\dot{\omega}_{\bar{y}} + \left(\Omega_{\bar{z}}\,\Theta a_{\bar{x}\bar{x}}\,\omega_{\bar{x}} - \Omega_{\bar{x}}\,\Theta a_{\bar{z}\bar{z}}\,\omega_{\bar{z}}\right) && = M_{a\bar{y}} \\
&\Theta a_{\bar{z}\bar{z}}\,\dot{\omega}_{\bar{z}} + \left(\Omega_{\bar{x}}\,\Theta a_{\bar{y}\bar{y}}\,\omega_{\bar{y}} - \Omega_{\bar{y}}\,\Theta a_{\bar{x}\bar{x}}\,\omega_{\bar{x}}\right) - ml\cos(\varphi)h\ddot{\varphi} && = M_{a\bar{z}}
\end{aligned}$$

$$\begin{aligned}
&\Theta a_{x\bar{x}}\,\dot{\omega}_{\bar{x}} - \left(\Omega_{\bar{z}}\,\Theta a_{\bar{y}\bar{y}}\,\omega_{\bar{y}} - \Theta a_{\bar{z}\bar{z}}\,\omega_{\bar{z}}\,\Omega_{\bar{y}}\right) && = M_{a\bar{x}} \\
&\Theta a_{\bar{y}\bar{y}}\,\dot{\omega}_{\bar{y}} - \left(\Omega_{\bar{x}}\,\Theta a_{\bar{z}\bar{z}}\,\omega_{\bar{z}} - \Omega_{\bar{z}}\,\Theta a_{\bar{x}\bar{x}}\,\omega_{\bar{x}}\right) && = M_{a\bar{y}} \\
&\Theta a_{\bar{z}\bar{z}}\,\dot{\omega}_{\bar{z}} - \left(\Omega_{\bar{y}}\,\Theta a_{\bar{x}\bar{x}}\,\omega_{\bar{x}} - \Omega_{\bar{x}}\,\Theta a_{\bar{y}\bar{y}}\,\omega_{\bar{y}}\right) - ml\cos(\varphi)h\ddot{\varphi} && = M_{a\bar{z}}
\end{aligned}$$

e)



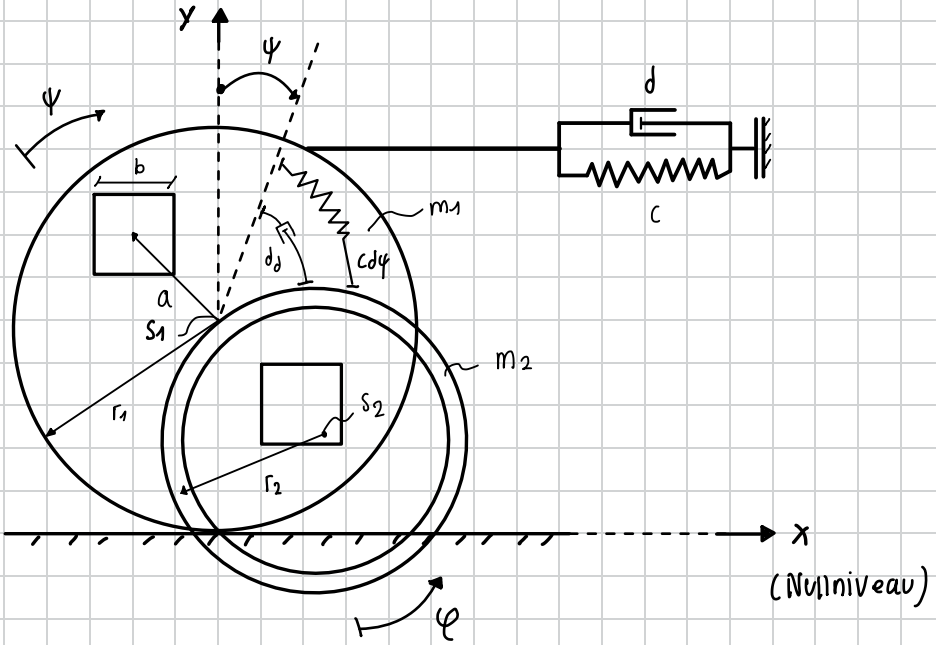




$$\Sigma M_a^x \stackrel{!}{=} 0: \quad -M_{S\bar{y}} = 0$$

$$\Sigma M_a^y \stackrel{!}{=} 0: \quad M_M - S_{\bar{z}}\cdot l = 0$$

$$\Sigma M_a^z \stackrel{!}{=} 0: \quad -M_{S\bar{z}} + S_{\bar{y}}\cdot l = 0$$

Aufg. 4 Lagrange 2. Art




a) Massenträgheitsmoment von $m_1$

b) $\underline{r}_{S1}$ und $\underline{r}_{S2}$ bestimmen, sowie $\dot{\underline{r}}_{S1}$, $\dot{\underline{r}}_{S2}$ und $\dot{\underline{r}}_{S1}^2$, $\dot{\underline{r}}_{S2}^2$

c) $E_{Kin}$ und $E_{Pot}$ bestimmen

d) $\varphi_\varphi^{enk}$ und $\varphi_\psi^{enk}$ bestimmen

e) Lagrange-Gleichungen in $\psi$ und $\varphi$ bestimmen

f) Linearisieren für $\varphi << 1$ und $\psi << 1$

g) } Schwingungsaufgaben (aus Lagrange Gleichungen)

h) }

b)

$$\underline{v}_{s_1} = \underline{v}_m + \begin{pmatrix} 0 \\ 0 \\ -\dot{\psi} \end{pmatrix} \times \begin{pmatrix} 0 \\ r_1 \\ 0 \end{pmatrix} = r_1 \dot{\psi}$$

$$\underline{r}_{s_1} = \begin{pmatrix} x_{s_1} \\ r_1 \\ 0 \end{pmatrix} = \begin{pmatrix} r_1 \psi \\ r_1 \\ 0 \end{pmatrix} \qquad \underline{r}_{s_2} = \begin{pmatrix} x_{s_1} + r_2 \sin(\psi) \\ r_1 - r_2 \cos(\psi) \\ 0 \end{pmatrix}$$

$$\underline{\dot{r}}_{s_1} = \begin{pmatrix} r_1 \dot{\psi} \\ 0 \\ 0 \end{pmatrix} \qquad \underline{\dot{r}}_{s_2} = \begin{pmatrix} r_1 \dot{\psi} + r_2 \dot{\psi} \cos(\psi) \\ r_2 \dot{\psi} \sin(\psi) \\ 0 \end{pmatrix}$$

$$\dot{r}_{s_1}^2 = r_1^2 \dot{\psi}^2$$

$$\dot{r}_{s_2}^2 = r_1^2 \dot{\psi}^2 + r_2^2 \dot{\psi}^2 \cos^2(\psi) + 2 r_1 \dot{\psi}^2 r_2 \cos(\psi) + r_2^2 \dot{\psi}^2 \sin^2(\psi)$$

$$\dot{r}_{s_2}^2 = r_1^2 \dot{\psi}^2 + r_2^2 \dot{\psi}^2 + 2 r_1 r_2 \dot{\psi}^2 \cos(\psi)$$

c)

$$T = T_1 + T_2$$

$$T_1 = \frac{1}{2} m_1 \dot{r}_{s_1}^2 + \frac{1}{2} \Theta_{s_1} \dot{\psi}^2 = \frac{1}{2} m_1 r_1^2 \dot{\psi}^2 + \frac{1}{2} \Theta_{s_1} \dot{\psi}^2 \quad ✓$$

$$T_2 = \frac{1}{2} m_2 \dot{r}_{s_2}^2 + \frac{1}{2} \Theta_{s_2} \dot{\varphi}^2 = \frac{1}{2} m_2 \left[ r_1^2 \dot{\psi}^2 + r_2^2 \dot{\psi}^2 + 2 r_1 r_2 \dot{\psi}^2 \cos(\psi) \right] + \frac{1}{2} m_2 r_2^2 \dot{\varphi}^2$$

$$T = \frac{1}{2} m_1 r_1^2 \dot{\psi}^2 + \frac{1}{2} \Theta_{s_1} \dot{\psi}^2 + \frac{1}{2} m_2 r_1^2 \dot{\psi}^2 + \frac{1}{2} m_2 r_2^2 \dot{\psi}^2 + m_2 r_1 r_2 \dot{\psi}^2 \cos(\psi) + \frac{1}{2} m_2 r_2^2 \dot{\varphi}^2$$

✓ ✓ ✓ ✓ ✓ ✓

$$U = U_1 + U_2 + U_c + U_{c_{d\psi}}$$

$$U_1 = 0$$

$$U_2 = m_2 g \left( r_1 - r_2 \cos(\psi) \right)$$

$$U_c = \frac{1}{2} c \left( 2 r_1 \psi + r_1 \sin(\psi) \right)^2$$

$$U_{c_{d\psi}} = \frac{1}{2} c_{d_\psi} (\psi + \varphi)^2$$

d)

$$Q_\varphi^{e,nk} = -d_d \cdot (\dot{\psi} + \dot{\varphi}) \cdot \frac{\partial \psi + \varphi}{\varphi} = -d \cdot (\dot{\psi} + \dot{\varphi})$$

$$Q_\psi^{e,nk} = M_m - d_d (\dot{\psi} + \dot{\varphi}) - d \cdot \left( 2 r_1 \dot{\psi} + r_1 \dot{\psi} \cos(\psi) \right) \cdot \frac{\partial 2 r_1 \psi + r_1 \sin(\psi)}{\partial \psi}$$

$$= M_m - d_d(\dot{\psi}+\dot{\varphi}) - d\cdot\left(2r_1\dot{\psi}+r_1\dot{\psi}\cos(\psi)\right)\cdot\left(2r_1+r_1\cos(\psi)\right)$$

e)

$$L = T - U$$

$$= \frac{1}{2}m_1 r_1^2\dot{\psi}^2 + \frac{1}{2}\Theta_{s_1}\dot{\psi}^2 + \frac{1}{2}m_2 r_1^2\dot{\psi}^2 + \frac{1}{2}m_2 r_2^2\dot{\psi}^2 + m_2 r_1 r_2\dot{\psi}^2\cos(\psi) + \frac{1}{2}m_2 r_2^2\dot{\varphi}^2$$
$$- m_2 g r_1 + m_2 g r_2\cos(\psi) - \frac{1}{2}c\left(2r_1\psi + r_1\sin(\psi)\right)^2 - \frac{1}{2}c_{d_\psi}(\psi+\varphi)^2$$